

Master's thesis



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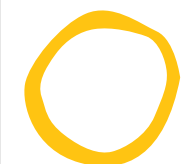
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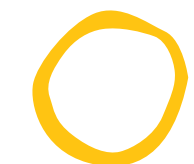
Major: Statistics and Data Science

**Topic: Prediction of Hotel Booking Cancellations Using Machine Learning
Techniques**

Year: 2026

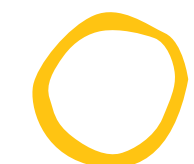
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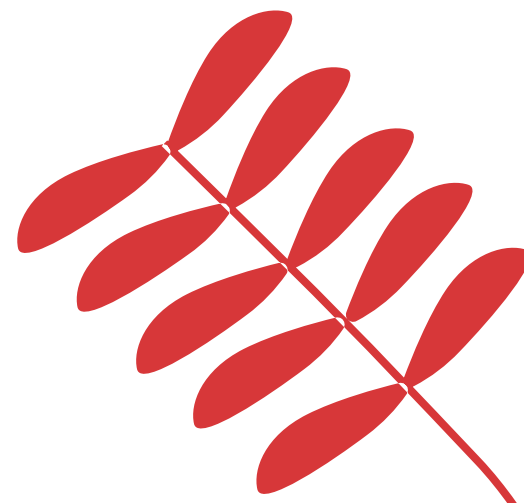
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INTRODUCTION

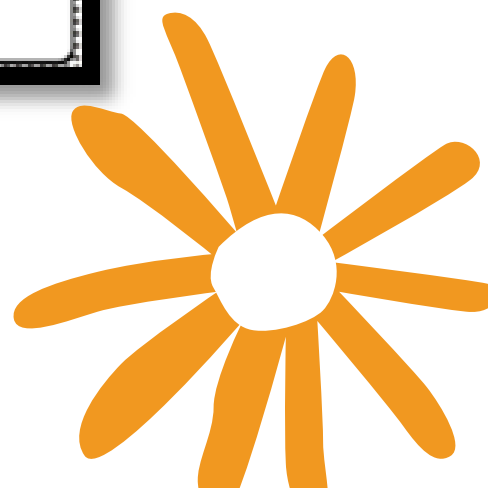
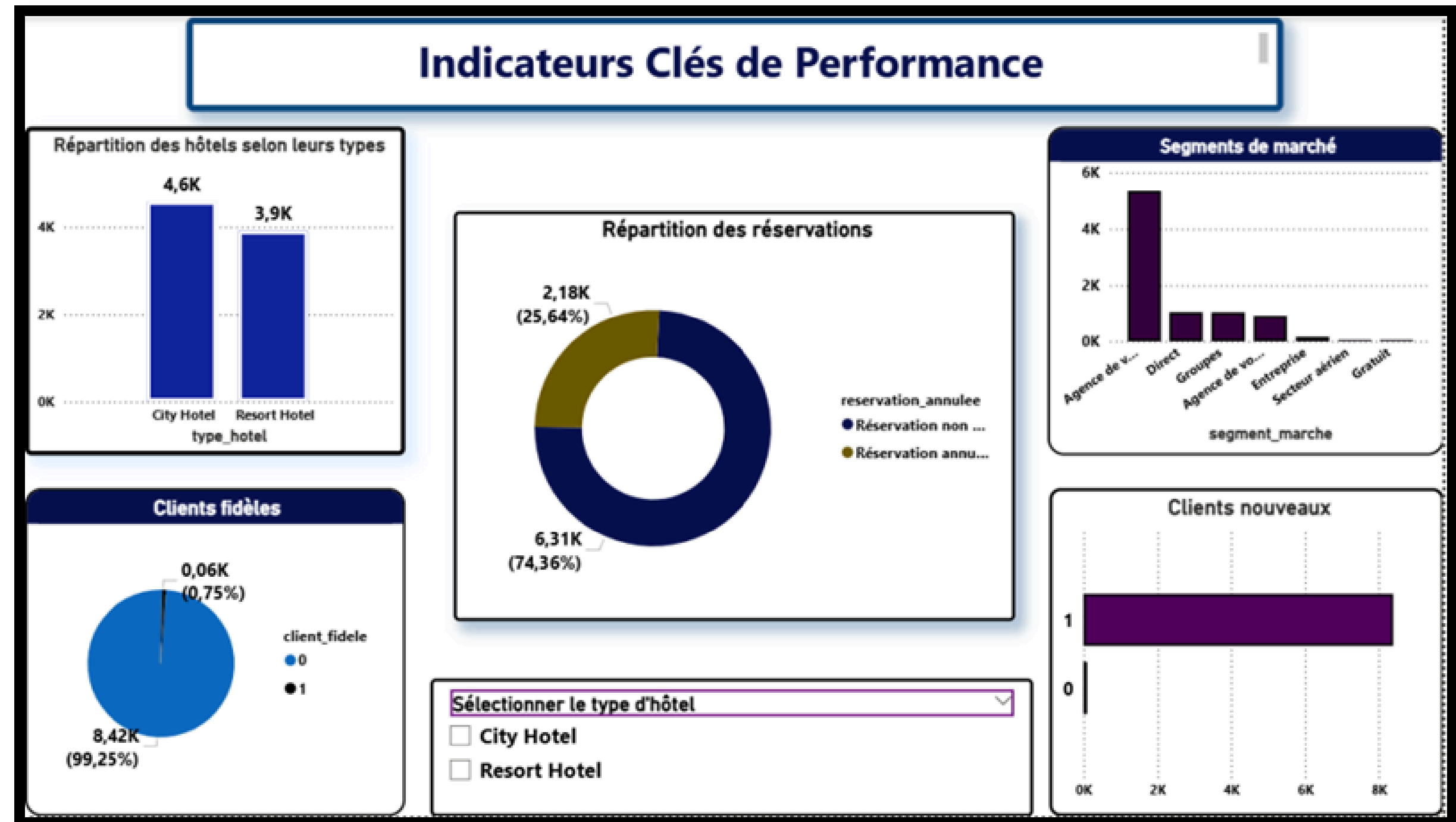
In this empirical section, we focus on the practical application of data analysis methods and Machine Learning to study the phenomenon of hotel booking cancellations. This is a binary supervised classification study whose main objective is to predict the binary target variable "canceled booking" (1 if canceled, 0 otherwise), while identifying the most influential factors driving this decision.

The adopted methodological approach is structured around four major steps:

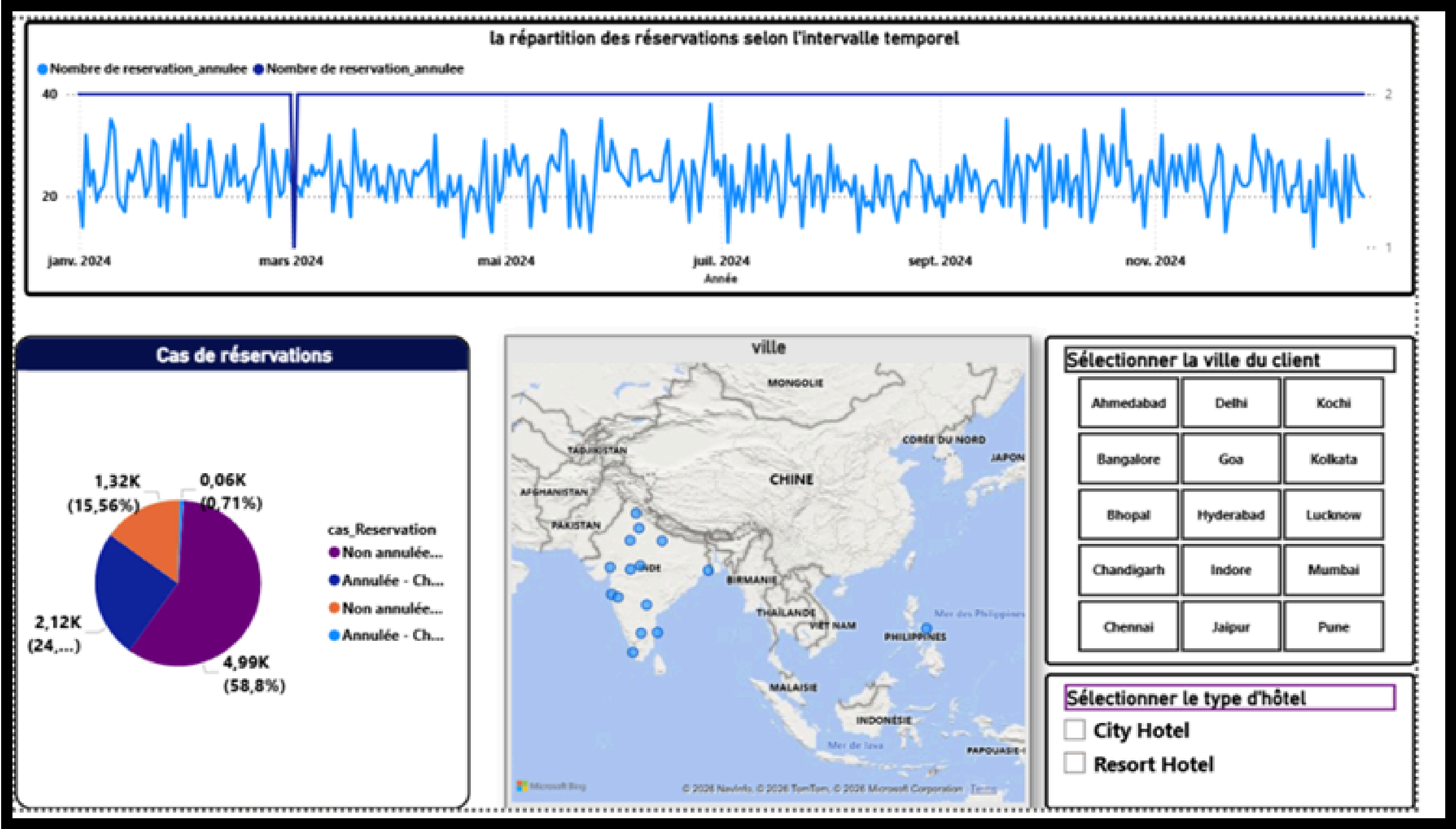
- 1. Data preparation and cleaning**
- 2. Exploratory Data Analysis (EDA)**
- 3. Feature selection and modeling**
- 4. Managerial recommendations**



DASHBOARDS

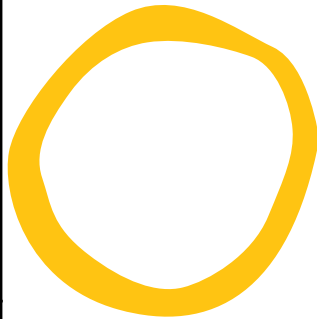
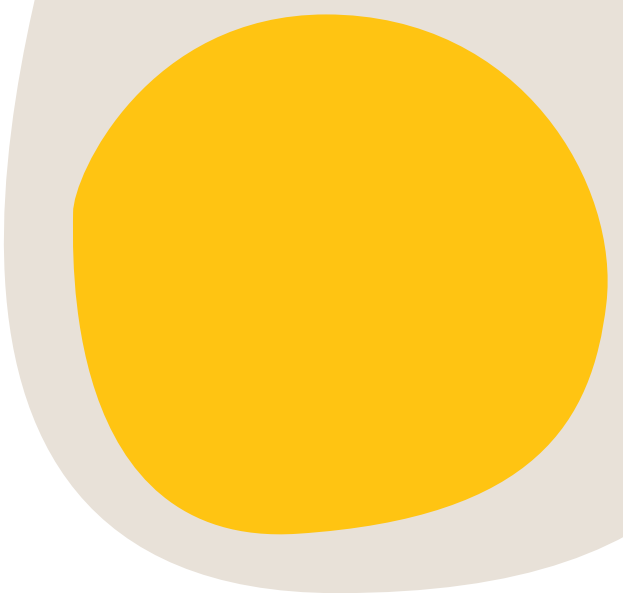


DASHBOARDS



RESULTS AND PERFORMANCE

Modèle	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Arbre de décision	82,5	65,05	66,12	65,58
Forêt aléatoire	85,63	73,83	66,58	70,02
SVM	82,8	70,48	54,67	61,57



RECOMMENDATIONS

- **STRENGTHEN DATA VALIDATION RULES TO PREVENT INCONSISTENCIES IN BOOKINGS AND CUSTOMER PROFILES.**
- **ENHANCE THE PLATFORM'S USER EXPERIENCE THROUGH AUTOMATED BOOKING CONFIRMATIONS AND USER GUIDES.**
- **IMPLEMENT LOYALTY PROGRAMS TO ENCOURAGE GUESTS TO HONOR THEIR RESERVATIONS.**
- **CONDUCT A DEEPER ANALYSIS OF SEGMENTS WITH ZERO CANCELLATIONS, PARTICULARLY GUESTS BOOKING ROOM TYPE I AND THOSE REQUESTING PARKING SPACES.**
- **ADOPT A PROGRESSIVE DEPOSIT POLICY TO REDUCE CANCELLATIONS FOR BOOKINGS MADE FAR IN ADVANCE.**
- **DEVELOP BUNDLED PACKAGES (COMBINING ACCOMMODATION WITH COMPLEMENTARY SERVICES) TO INCREASE CUSTOMER COMMITMENT.**
- **TIGHTEN COLLABORATION TERMS WITH ONLINE TRAVEL AGENCIES (OTAS) BY ENFORCING STRICTER CANCELLATION POLICIES AND UPFRONT PAYMENTS.**



CONCLUSION

BY COMBINING THE PREDICTIVE EFFICIENCY OF THE RANDOM FOREST MODEL WITH THESE TARGETED PRICING AND CUSTOMER-RELATIONSHIP POLICIES, THE ESTABLISHMENT ENSURES A SIGNIFICANT REDUCTION IN UNEXPECTED CANCELLATIONS, THEREBY OPTIMIZING CAPACITY MANAGEMENT AND OVERALL PROFITABILITY.





**THANK YOU
FOR YOUR
ATTENTION**

